

DUAL PORPHYRIAS

In the dual porphyrias, laboratory findings may indicate a simultaneous deficiency of two enzymes in the heme biosynthetic pathway, occurring either in a single individual or within a family. To date, approximately 15 patients and families with such presentations have been reported, highlighting the extreme rarity of these variants.

Most cases of dual porphyria involve a deficiency of **uroporphyrinogen (UROGEN) decarboxylase** combined with a deficiency of **porphobilinogen (PBG) deaminase**, **coproporphyrinogen (COPROGEN) oxidase**, or **protoporphyrinogen (PROTOGEN) oxidase**. This is not unexpected, given that **porphyria cutanea tarda (PCT)**—associated with UROGEN decarboxylase deficiency—is the most prevalent form of human porphyria. In three reported families, concurrent deficiencies of **COPROGEN oxidase** and either **PBG deaminase**, **UROGEN III synthase**, or **ALA dehydratase** were identified. One patient with **congenital erythropoietic porphyria (CEP)** exhibited a UROGEN III synthase defect along with a deficiency of UROGEN decarboxylase.

Historically, the diagnosis of dual porphyria has relied on biochemical measurements of porphyrins and porphyrin precursors in urine, feces, and plasma, as well as enzymatic assays in erythrocytes, lymphocytes, or fibroblasts.

Recently, two independent studies have used DNA analysis to definitively confirm dual porphyria by identifying mutations in two distinct genes encoding enzymes in the heme biosynthesis pathway.

- **Harraway et al.** described a young female patient with photosensitive skin lesions and increased urinary excretion of PBG. Initial suspicion of **variegate porphyria (VP)** was not confirmed. Biochemical profiling of urine, feces, and plasma instead suggested combined deficiencies of **PBG deaminase** and **UROGEN decarboxylase**. Automated DNA sequencing revealed pathogenic mutations in both the *HMBS* (PBG deaminase) and

UROD (UROGEN decarboxylase) genes, confirming the diagnosis of dual porphyria.

- **Akagi et al.** reported a patient with acute neurovisceral attacks and elevated urinary levels of **ALA**, **PBG**, and **coproporphyrin (COPRO)**—findings suggestive of **hereditary coproporphyria (HCP)**. However, markedly increased **erythrocyte zinc protoporphyrin** levels also pointed to **ALA dehydratase deficiency**. DNA sequencing identified disease-causing mutations in both the *CPOX* (COPROGEN oxidase) and *ALAD* (ALA dehydratase) genes. Functional expression studies confirmed both defects, establishing a molecular diagnosis of dual porphyria due to concurrent **COPROGEN oxidase** and **ALA dehydratase** deficiencies.

These cases underscore the importance of molecular genetic analysis in patients where biochemical and enzymatic data suggest dual enzyme deficiencies. Only through DNA sequencing and functional validation can the precise genetic basis of dual porphyria be established, enabling accurate diagnosis and informed genetic counseling.

ABNORMAL PORPHYRIN PROFILE AND SIDEROBLASTIC ANEMIA

Some forms of **sideroblastic anemia** are associated with abnormal porphyrin metabolism. In these cases, impaired heme synthesis leads to accumulation of iron in mitochondria (ringed sideroblasts) and may be accompanied by elevated levels of specific porphyrins, particularly **protoporphyrin**. This overlap reflects the shared pathway between heme biosynthesis and mitochondrial iron utilization. Differentiating between inherited or acquired sideroblastic anemias and porphyrias requires careful integration of clinical, hematologic, and biochemical data, and in some instances, molecular genetic testing.